

Circuit and working

As can be seen from Fig. 2, inputs to the accumulator, B-register, output register, PC register and memory address register are fed through the 8-bit parallel address/data bus. The input to the ALU is provided by the accumulator and B-register.

The ALU outputs to the bus via tristate buffers. The accumulator also outputs to the bus via tristate buffer 74LS245N. The tristate buffers are controlled by enable (G) pin.

The PC register output is fed to the program counter, providing the value to be loaded in the counter. The PC outputs to the bus via a tristate buffer. The memory address register outputs to the multiplexer.

The multiplexer output is fed to address pins of the RAM (CY6264). Input/output pins of the RAM are connected to the bus via the tristate buffer (IC23). The DIR and G pins of the tristate buffer are controlled based on the instruction and data flow. The 4-bit LSB of the bus is fed to the instruction register (74LS173N).

The inputs to the counter (CD4029N) are grounded. The outputs of the instruction register and the counter are connected to the address pins of ROMs (AT28C64). The ROM outputs are connected to enable pins of various ICs to control the data flow based on the state and instruction.

The functions of different modules in the architecture are as follows:

Program counter. The program counter provides the computer with the current address of the instruction to be executed.

Memory address register. The memory address register (MAR) stores the current address for the byte to be read or written from the memory.

Multiplexer. The multiplexer allows selection of address either from the MAR (in run mode) or the manual input (programming mode).

TABLE I
ROM CONTROL PINS

ROM1		ROM2	
D0	Program counter clock enable	D0	Instruction register reset
D1	Program counter tristate enable	D1	ALU decoder pin 1
D2	Memory address register clock enable	D2	ALU decoder pin 2
D3	Instruction register clock enable	D3	ALU decoder pin 3
D4	RAM tristate enable	D4	PC load register clock enable
D5	Accumulator clock enable	D5	RAM tristate data direction select
D6	Accumulator tristate enable	D6	Program counter load data
D7	B-register clock enable	D7	Output register clock enable

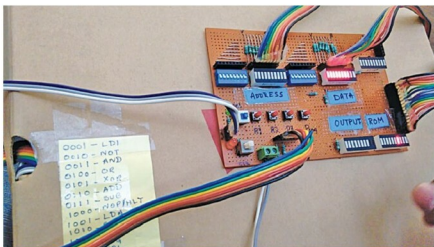


Fig. 3: Author's prototype

Instruction register. The instruction register of a computer stores the current instruction that is being executed.

During the computer's operation, the value stored in instruction register is the instruction opcode. It forms the four bits of middle-value address (A4-A7) for ROMs. The counter for the next state provides the four bits of LSB (A0-A3) for ROMs. The 3-bit output from the comparator (74LS85N) forms the MSB (A8-A10). Thus, the address (A0-A10) is fed to the two ROMs to generate the control logic for different modules.

Output register. It stores the value of the accumulator and displays the result.

Clock. The clock pulse is generated by an NE555 timer chip configured in astable mode.

ROM control

The two ROMs along with a 3-to-8 line decoder (74LS42N) provide a total of 18 control lines that control, enable or reset different parts of the computer. These control the clock enable pins of chips, thus controlling the data/instruction flow between various parts of the computer. The decoder output controls the tristates of the ALU controlling the data flow of the ALU result into the bus.

The ROM address is fed by the instruction register and counter. The data against the address for each state in the ROM is provided through the control lines or input/output pins. The functions of ROM control (I/O) pins, as per the schematic in Fig. 2, are listed in Table I.